

SlickTrace

Real-time oil-spill detection and vessel attribution - Gulf of Finland

Technical documentation - version 1.0 - 25 August 2026

1 Overview

SlickTrace watches a sea area continuously and turns public satellite, weather and ship-transponder feeds into a ranked list of suspect vessels for an oil discharge. The area of interest (AOI) is the Gulf of Finland, bounding box 21.8E 58.9N to 30.6E 60.7N. The system polls live AIS traffic from Digitraffic, hourly wind, wave and current fields from Open-Meteo, and Sentinel-1 GRD radar scenes from the Copernicus Data Space Ecosystem (CDSE). Each radar scene is scanned for dark patches; candidates are scored for oil-likeness; accepted slicks are hindcast backwards to an estimated release point and time; and every vessel recorded near that point around that time is ranked by weighted evidence.

Two constraints shaped the design. First, the system runs on real data only; there is no synthetic input anywhere. Second, detection had to work before any labelled data existed, so the oil classifier ships with a physics-based prior and treats trained models as optional upgrades.

The product is a single-page operations console served by the backend itself: a dark chart with live traffic, slick polygons, hindcast paths and forecast cones, plus panels for events, detections, satellite scenes and the spill-risk model.

2 Repository layout

```
backend/  
  main.py          FastAPI app: REST routes, WebSocket endpoint, static frontend  
  scheduler.py     System class: background loops, WebSocket hub, analysis flow  
  config.py        .env settings and physics constants  
  store.py         SQLite schema and access layer  
  vessel_info.py   vessel card data: MMSI flag lookup, ship types, computed history  
  providers/  
  detection/       ais_digitraffic.py, sentinel_cdse.py, weather_openmeteo.py  
                  sar_preprocess.py, darkspot.py, unet.py, features.py,  
                  classify.py, characterize.py, pipeline.py, train_detector.py  
  drift/           fields.py (interpolation), lagrangian.py (particle model)  
  attribution/  
  risk/            candidates.py, score.py, behavior.py  
                  features.py (grid feature engineering)  
frontend/src/  
scripts/  
  build_unet_dataset, train_unet, benchmark_segmenters, inspect_state  
tests/  
  test_drift.py, test_attribution.py  
data/  
  app.db, landmask/, scenes/, risk_model.joblib, unet_ds/, training/
```

3 Technology stack

Layer	Choice
Backend language	Python 3.11+
Web framework	FastAPI with uvicorn; REST plus a WebSocket channel
Database	SQLite (single file data/app.db), thread-locked access layer
Geospatial	rasterio (WarpedVRT), shapely, pyproj, pyshp, scipy.ndimage
Machine learning	scikit-learn (RandomForest), PyTorch (U-Net), joblib
Numerics	NumPy, SciPy, pandas
HTTP client	httpx (async), websockets

Layer	Choice
Frontend	React 18, Vite 5, Leaflet 1.9 with CARTO dark basemap; built bundle served by FastAPI as a single-page app
Configuration	python-dotenv (.env)

4 Runtime architecture

On startup `main.py` creates a `System` object (`scheduler.py`) that owns three background loops and a `WebSocket` hub:

- AIS loop: polls the Digitraffic snapshot every 30 seconds, inserts new positions into SQLite, refreshes vessel metadata every 10 minutes, keeps an in-memory dictionary of latest positions for cheap UI queries, prunes position history older than 7 days.
- Met-ocean loop: rebuilds the forcing grids every hour (past 4 days plus 3-day forecast) and republishes a `FieldSet` used by the drift model.
- Satellite loop: searches the CDSE catalogue every 10 minutes for Sentinel-1 GRD scenes intersecting the AOI in the last 48 hours, stores footprints, and when credentials are configured automatically downloads and scans the newest scene.

All events (new scenes, download progress, detection results, alerts, completed analyses) are broadcast as JSON over `WebSocket` `/ws` to connected dashboards. Heavy CPU work (archive unpacking, calibration, segmentation, particle integration, scoring) runs in worker executors so the event loop and the live feed never block.

5 External data feeds

Feed	Provider and endpoint	Notes
AIS positions	Digitraffic (Finnish Transport Infrastructure Agency), <code>meri.digitraffic.fi/api/ais/v1/locations</code>	No key. GeoJSON snapshot, polled every 30 s. <code>timestampExternal</code> is ms since epoch. Positions outside the AOI are dropped. Licence CC-BY 4.0.
Vessel metadata	<code>meri.digitraffic.fi/api/ais/v1/vessels</code>	Name, ship type, IMO, call sign, dimensions, draught, destination; refreshed every 10 min.
Wind	Open-Meteo forecast API	10 m wind speed and direction, converted to u/v components, hourly.
Currents and waves	Open-Meteo marine API	Ocean-current velocity (converted from km/h) and direction, wave height, direction, period. <code>past_days=4</code> , <code>forecast_days=3</code> .
Radar scenes	Copernicus Data Space Ecosystem OData catalogue	Catalogue search is keyless and returns <code>GeoFootprint</code> polygons. Download requires a free account; OAuth2 password grant with <code>client_id</code> <code>cdse-public</code> . The catalogue answers with a 301 redirect to <code>download.dataspace.copernicus.eu</code> ; the client follows redirects manually so the bearer token is re-attached.
Coastline	Natural Earth 10 m land shapefile	Rasterised onto any SAR grid for land masking and shore-distance fields.

Feed	Provider and endpoint	Notes
Historical slicks	SkyTruth Cerulean OGC API, collection public.slick_plus	ML detections of slicks in Sentinel-1 imagery since 2020 over the Baltic (bbox 9,53 to 31,66). Anthropogenic classes 2, 4, 5, 6, 7, 8. Used as training corpus for the risk model and as masks for the U-Net.

Met-ocean forcing is sampled on a 6 x 5 point grid across the AOI and stored as arrays of shape [lat, lon, hour]. The drift model reads them through bilinear spatial and linear temporal interpolation.

6 SAR detection pipeline

Entry point: `DetectionPipeline.process_product` (backend/detection/pipeline.py). Stages in order:

6.1 Preprocessing

- SAFE archive unpacking; measurement GeoTIFF selection prefers the VV polarisation because oil contrast is stronger and less wind-saturated than VH at Baltic wind speeds.
- Calibration: DN values are squared and divided by the sigma-nought LUT parsed from the product calibration XML, indexed by azimuth line and range pixel, then converted to decibels.
- Geolocation: the measurement TIFF carries ground control points rather than an affine transform, so reading goes through a rasterio WarpedVRT warp to EPSG:4326. Large windows are decimated to stay under 40 megapixels.
- Speckle filtering: Lee filter on the dB image, NaN-safe, 7 x 7 window.
- Land masking: Natural Earth polygons rasterised onto the image grid; sea pixels only proceed.

6.2 Segmentation

Detector selection is controlled by `DETECTOR=auto|unet|heuristic` in `.env`. With `auto`, the trained U-Net is used when `data/unet_model.pt` exists, otherwise the heuristic segmenter runs.

Heuristic segmenter (backend/detection/darkspot.py). Oil films damp capillary and short gravity waves, so slicks appear as dark sharp-edged patches. The segmenter computes per-block (512 px) sea background statistics and thresholds at mean minus 2.2 standard deviations; blocks without enough valid sea fall back to global statistics. Binary closing removes speckle; components between 0.05 and 80 square kilometres are kept.

U-Net (backend/detection/unet.py). A compact encoder-decoder: one input channel (dB), base width 16, three pooling levels with skip connections, batch normalisation, 1 x 1 output convolution producing an oil logit. Input tiles are 256 x 256, dB range -32 to -8 normalised to 0..1 with no-data mapped to neutral sea level. Inference tiles large images with plain stitched 256 px windows.

Training data is real and paired: Sentinel-1 imagery downloaded from CDSE for the Baltic scenes with the most Cerulean slicks, masked with Cerulean anthropogenic polygons of machine confidence 0.70 or higher from the same scenes. Tiles with at least 40 positive pixels are positives; negatives are subsampled at a 1.5:1 ratio. The split is strictly by scene (12 train, 2 validation, 2 test) so no scene contributes to two splits. Loss is binary cross-entropy plus soft Dice; optimiser Adam at learning rate 0.001; augmentation is rotation by multiples of 90 degrees and horizontal flip. The best checkpoint by validation IoU is kept.

`scripts/benchmark_segmenters.py` compares both segmenters on the held-out scenes against the Cerulean masks, reporting pixel IoU, tile-level detection recall and false alarms per tile.

6.3 Patch features

Each candidate patch gets physics-meaningful measurements (backend/detection/features.py):

- area in pixels and square kilometres; complexity, defined as perimeter divided by the perimeter of the equal-area circle;

- contrast against the surrounding sea ring, in dB, computed on a dilated ring minus the patch itself;
- edge sharpness, the mean gradient magnitude in a one-pixel boundary band;
- GLCM homogeneity on a 16-level quantised crop with offset 3 pixels;
- elongation and orientation from covariance eigenvalues of patch pixels;
- local dark fraction in a window four times the patch extent (oil sits alone in clean water; wind cells come in crowded fields);
- distance to nearest land pixel from a Euclidean distance transform.

6.4 Oil versus look-alike scoring

Two paths exist in `classify.py`. The `RandomForest` path activates automatically when `data/model_rf.joblib` exists (train it with `scripts/train_detector.py` after dropping labelled crops into `data/training/oil` and `data/training/lookalike`). Otherwise a physics-informed prior produces the score:

Prior term	Weight	Behaviour encoded
Contrast	0.30	strong damping of backscatter is the primary signature
Size	0.20	log-scale saturating term; larger patches score higher
Edge sharpness	0.15	man-made films have crisp boundaries
Homogeneity	0.10	mineral oil is smoother than biogenic filaments
Complexity	0.10	compact or smoothly elongated beats ragged
Isolation	0.05	clean water around the patch
Distance from land	0.10	coastal look-alikes and wind shadows penalised

Accepted patches are vectorised to simplified lon/lat polygons and stored with centroid, area, major and minor axis, orientation, and the score as both confidence and `oil_probability`. Deduplication is per scene only; linking the same slick across passes would require drift correlation and is deliberately not faked.

7 Drift model

The model (`backend/drift/lagrangian.py`) advects a Lagrangian particle ensemble in a local tangent-plane frame in metres, anchored at the slick centroid. Transport per particle is the sum of:

- the interpolated surface current;
- windage leeway: 3 percent of the 10 m wind speed, deflected 20 degrees to the right of the wind (northern-hemisphere Ekman/leeway approximation);
- a Stokes drift approximation: 2 percent of the wind speed, aligned with the wind;
- random-walk diffusion applied to positions (Euler-Maruyama) with eddy diffusivity 1.5 square metres per second.

Integration is second-order Runge-Kutta (midpoint) with a 300 second step and 600 particles seeded uniformly inside the slick polygon.

Backward run (default 18 hours). Time is reversed and diffusivity is damped to 35 percent to limit artificial ensemble inflation during backtracking. Ensemble spread (root mean square distance to the centroid) is recorded hourly. The spread curve contracts toward the source; its minimum region marks the release. The implementation takes the earliest sample within 15 percent of the global minimum, which biases the estimate toward a younger age. The chosen sample gives the estimated release point, release time, slick age, and an origin uncertainty taken from the ensemble spread at that horizon (at least 0.5 km).

Forward run (default 24 hours). Same dynamics integrated forward; output is the centroid path plus hourly uncertainty circles (cone radii) used by responders and drawn on the chart.

8 Vessel attribution

For a given origin estimate, candidate selection (backend/attribution/candidates.py) collects every MMSI with at least one fix within radius $\max(3 \times \sigma + 4, 12)$ kilometres of the release point in the window from 8 hours before to 2.5 hours after the estimated release moment. Vessels with fewer than two fixes are skipped. Each remaining vessel receives seven evidence factors, combined as a weighted sum and scaled to 0..100 (backend/attribution/score.py):

Factor	Weight	Definition
proximity	0.33	closest approach to the release point among fixes within 45 minutes of the release time (fallback: closest fix in the window); $\exp(-d / 4 \text{ km})$
crossing	0.13	track passes within 2 km of the origin point scores 1.0; intersection with the slick footprint (buffered about 1 km) scores 0.7; otherwise 0
speed_anomaly	0.13	fraction of fixes below 2 knots, boosted when the median speed over the window is below 5 knots (drifting or loitering)
ais_gap	0.13	longest AIS silence exceeding 20 minutes that overlaps the release moment with a 30 minute tolerance; value saturates at 2 hours of silence (dark event)
type_prior	0.09	static prior by AIS ship type: tankers (codes 80 to 89) 1.00, cargo 0.75, tug 0.55, towing and port tender 0.45, reserve 0.40, fishing 0.25, pilot 0.30, law enforcement 0.35, passenger 0.20, unknown 0.35
course_align	0.11	net displacement bearing over the window versus the slick elongation axis; linear falloff to zero beyond 35 degrees
behavior_prior	0.08	learned anomaly score from the accumulated AIS history, see 8.1

Every factor carries a human-readable evidence string, shown in the dashboard (for example: closest approach 1.4 km, 12 min from est. release; AIS silent 96 min overlapping release window). The top 15 candidates are persisted per slick with their factors.

8.1 Vessel-behaviour prior

backend/attribution/behavior.py computes, per MMSI, indicators associated with illegal discharge behaviour in the research literature, from the watch's own AIS history (minimum 20 fixes, result cached 10 minutes):

- fraction of fixes below 1.5 knots more than 3 km offshore (slow in open sea);
- fraction of slow fixes during 22:00 to 05:00 UTC in open sea;
- longest gap between consecutive fixes.

These combine as $0.45 \times \text{slow_open_sea} + 0.30 \times \min(2 \times \text{night_slow}, 1) + 0.25 \times \min(\text{max_gap_min} / 240, 1)$. This is a risk prior, not an accusation; fishing, pilotage and port work also produce low speeds, which is why the factor weight is the smallest of the seven.

8.2 Vessel card

Clicking a vessel opens a card assembled from three sources: Digitraffic metadata; a flag country derived from the ITU Maritime Identification Digits table (first three digits of the MMSI); and computed history from the local store: total and 24-hour distance sailed, average and maximum speed, stops of 15 minutes or more below 1 knot (top eight with coordinates), count of dark gaps above 30 minutes, and the last 240 speed reports as a sparkline.

9 Spill-risk prediction model

A RandomForest predicts which sea cells are likely to accumulate anthropogenic slicks before a slick exists. Corpus: historical slick detections from SkyTruth Cerulean (public.slick_plus collection, 2020 to 2026, Baltic bounding box), roughly 40,000 records of which about 2,500 anthropogenic detections fall inside the AOI across 865 distinct satellite scene visits. Target: 0.05 degree (about 5 km) grid cells containing at least one anthropogenic slick.

Features per sea cell (backend/risk/features.py): distance to shore (Natural Earth rasterised, then a distance transform), logarithm of accumulated AIS fix density, distance to the nearest high-density traffic lane (top decile of density cells, then a distance transform), distance to the nearest of 13 major Baltic oil terminals (Primorsk, Ust-Luga, Vysotsk, Saint Petersburg, Sillamae, Tallinn-Muuga, Kilpilahti, Helsinki Vuosaari, Naantali, Klaipeda, Gdansk, Riga, Nynashamn), plus cell latitude and longitude.

Training protocol (scripts/train_risk_model.py): 300-tree forests per fold, validated with spatially grouped 5-fold cross-validation where groups are one-degree longitude bands. Random cell splits would overstate performance because neighbouring cells share terminals and lanes; grouped folds measure generalisation to unseen sea areas. Mean held-out ROC AUC is about 0.75. The final model (400 trees, min_samples_leaf 5, balanced_subsample class weighting) is fitted on all cells and saved with metadata (feature importances, per-fold AUC, corpus description) to data/risk_model.joblib; per-cell probabilities are written to the risk_grid table and served as GeoJSON.

Rebuild commands: python scripts/fetch_cerulean_slicks.py, then python scripts/train_risk_model.py.

10 Web dashboard

Single-page React application (frontend/src), built by Vite into frontend/dist and served by FastAPI. Layout:

- Header: SLICKTRACE brand, three feed LEDs (AIS, MET-OCEAN, SENTINEL-1) driven by status freshness, a RISK LAYER toggle, and a UTC clock.
- Left panel with four tabs: EVENTS (alert feed), SLICKS (detections with area, confidence and estimated age), SCENES (catalogue queue; clicking a row downloads and scans that scene; status chips cover downloading, detecting, slick candidates, sea clear, error), RISK (model card with corpus size, AUC and top feature importances, plus the ten highest-risk cells which fly the map to them).
- Map: CARTO dark basemap; cyan dots for live AIS (clickable, tooltip with name and speed); red slick polygons; amber dashed hindcast path to a pulsing origin crosshair sized by uncertainty; cyan dashed forecast path with uncertainty circles; optional amber risk heatmap with legend; hydrographic coordinate readout under the cursor; selected-vessel tracks drawn on demand.
- Right panel: slick detail (properties, release estimate, ranked suspect list with score bars and expandable per-factor evidence bars, re-analyse button) or the vessel card described in section 8.2.

The app polls REST every 15 seconds and applies WebSocket messages immediately for event, scene and analysis updates.

11 HTTP API and WebSocket

Endpoint	Purpose
GET /api/status	pipeline health: feed ages and errors, stored position count, live vessel count, met field coverage, scene status counts, active download progress
GET /api/vessels/live	GeoJSON FeatureCollection of current AIS positions in the AOI
GET /api/vessels/{mmsi}/track?hours=	one vessel's track history
GET /api/vessels/{mmsi}/details	vessel card payload (metadata, flag, computed history)
GET /api/scenes	Sentinel-1 catalogue cache, newest first
POST /api/scenes/{id}/scan	download a scene and run detection
GET /api/slicks	all detected slicks with geometry
GET /api/slicks/{id}	slick detail plus backward and forward drift runs and ranked suspects
POST /api/slicks/{id}/analyze	re-run hindcast, forecast and attribution

Endpoint	Purpose
GET /api/events	alert and event log
GET /api/risk/status	risk model card: AUC, fold AUCs, importances, corpus size
GET /api/risk/grid?min_p=	risk probability cells as GeoJSON polygons
WS /ws	live push: hello/status on connect, ais_tick, scene, scene_status, event, analysis_complete

12 Database schema

Table	Contents
ais_positions	mmsi, epoch ts, lon, lat, sog, cog, navstat; indexed on (mmsi, ts) and ts; pruned to 7 days
vessels	one row per MMSI: name, ship_type, destination, draught, imo, call_sign, length, width
scenes	product_id, name, sensed_start, size_mb, footprint, status (catalogued, downloading, processing, detected, clear, error), error text
slicks	scene reference, detected_at, centroid, area_km2, axes, orientation, confidence, oil_probability, age_estimate_h, age_sigma_h, GeoJSON geometry, properties JSON
drift_runs	one row per run per direction: origin_lon/lat, origin_sigma_km, release_time, spread_curve, centroid path, forward cone
suspects	slick_id, mmsi, score, rank, factors JSON ({name: {score, weight, evidence}}), unique per (slick_id, mmsi)
events	timestamped log entries with kind, severity, message, payload
risk_grid	risk-model probabilities per grid cell (lon0, lat0, lon1, lat1, p)
hist_slicks	created by scripts/fetch_cerulean_slicks.py; Cerulean corpus: id, ts, class, confidence, area, length, centroid, source scene id

13 Scripts

Script	What it does
scripts/fetch_landmask.py	downloads and extracts the Natural Earth 10 m land shapefile
scripts/fetch_cerulean_slicks.py	pages through the Cerulean OGC API (1000 items per page) into hist_slicks
scripts/train_risk_model.py	builds features, trains the risk RandomForest with spatially grouped CV, writes risk_model.joblib and the risk_grid table
scripts/build_unet_dataset.py	selects Baltic scenes with the most Cerulean slicks, downloads them from CDSE, extracts paired 256 px image/mask tiles, assigns train/val/test splits by scene, deletes archives afterwards to save disk
scripts/train_unet.py	trains the segmenter (BCE + Dice, Adam, augmentation), saves the best checkpoint by validation IoU to data/unet_model.pt
scripts/benchmark_segmenters.py	evaluates U-Net versus heuristic on held-out test scenes: pixel IoU, detection recall, false alarms
scripts/train_detector.py	optional: trains the patch-level RandomForest from crops in data/training/{oil,lookalike}, saves data/model_rf.joblib

Script	What it does
scripts/inspect_state.py	prints slick, suspect and score summaries from the database

14 Tests

Run with: `python -m pytest tests/ -q`

- test_drift.py: interpolation returns exact uniform-field values; local frame round-trip is exact; a 6-hour forward run under a 1 m/s eastward current moves the ensemble centroid at 3.2 to 4.8 km/h (current plus windage plus Stokes); a 10-hour backward run places the estimated origin within 1 degree of the seed under zero net transport.
- test_attribution.py: a stationary tanker sitting at the origin outranks a fast transiting cargo several kilometres away; a vessel with a 3-hour AIS silence over the release window outranks an equivalent clean track; all composite scores stay within 0 to 100.

15 Configuration (.env)

Key	Default	Meaning
CDSE_USER / CDSE_PASS	empty	Copernicus Data Space account; needed for scene download
AOI_BBOX	21.8,58.9,30.6,60.7	area of interest, lon0,lat0,lon1,lat1
AIS_POLL_SECONDS	30	Digitraffic polling interval
MET_REFRESH_SECONDS	3600	met-ocean grid refresh interval
SAT_POLL_SECONDS	600	sentinel catalogue poll interval
DRIFT_HOURS_BACK	18	hindcast horizon
DRIFT_HOURS_FWD	24	forecast horizon
PARTICLES	600	particles per drift run
DETECTOR	auto	unet heuristic auto (auto prefers the trained U-Net)
DATA_DIR / DB_PATH	data / data/app.db	storage locations

16 Running the system

```

pip install -r requirements.txt
copy .env.example .env    (add CDSE_USER / CDSE_PASS)
cd frontend && npm install && npm run build && cd .. (first run only)
run.bat    (or: python -m uvicorn backend.main:app --port 8000)

```

Open <http://localhost:8000>. The AIS LED turns green within seconds (hundreds of live vessels in the Gulf), met-ocean fields cache within a minute, and the scene list fills from the 48-hour catalogue. Click any scene to download and scan it; detected slicks with confidence 0.55 or higher are analysed automatically and appear under SLICKS with origins and suspect rankings.

17 Artifacts currently on disk

- data/app.db: populated store (live AIS history, catalogued and processed scenes, detected slicks, drift runs, suspects, events, risk grid, Cerulean corpus).
- data/risk_model.joblib: trained risk model with metadata.

- data/unet_model.pt (if trained): U-Net checkpoint with validation IoU; used automatically by DETECTOR=auto.
- data/scenes/: unpacked SAFE products, including two full Sentinel-1 GRD scenes from November 2023 and April 2024 retained from dataset work.
- data/landmask/: Natural Earth 10 m shapefile.
- data/unet_ds/, data/training/: optional training corpora.

18 Known limitations

- Attribution is probabilistic evidence ranking, not proof; enforcement requires sample collection, bunker delivery records and investigation.
- AIS gaps also arise from coastal coverage shadowing rather than deliberate silence, which is why the gap factor weight is modest.
- Open-Meteo currents are about 8 km resolution: adequate for Gulf-scale hindcasts, coarse in coastal pockets.
- The day-1 detector prior is calibrated for C-band IW GRD conditions in the Baltic; other basins need retraining.
- Risk-grid probabilities shown on the map are in-sample; the honest generalisation figure is the spatially grouped cross-validation AUC.
- The same slick seen in consecutive passes is not yet linked; deduplication is per scene only.

19 Data source credits

- Copernicus Data Space Ecosystem: Sentinel-1 GRD imagery.
- Fintraffic / Digitraffic: AIS (CC-BY 4.0).
- Open-Meteo: wind, marine and current fields (ECMWF/CMEMS-derived).
- SkyTruth Cerulean: historical slick detections and segmentation masks.
- Natural Earth: coastlines.